

AI-Enhanced Methodologies and Agents for Business Analysis in Software Development

Adaptive Prompt Engineering, Ethical Prompting, and Hybrid AI Agents
for Business Analysis

Research Proposal

CSC4046 – Individual Research Project

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Declaration

I hereby declare that this research will be conducted by me and will be my own effort, with no part plagiarized and with proper citations. This research proposal is submitted in partial fulfillment of the requirements for the course unit **CSC4046 — Individual Research Project**, under the supervision of **Dr. S.W.A.M. Upamalika**.

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Chapter 1

Introduction

1.1 Introduction

Business Analysts (BAs) bridge stakeholder needs and technical implementation across the SDLC by eliciting, analyzing, and documenting requirements. However, BA work is often slowed by ambiguity, frequent changes, and communication gaps between stakeholders and delivery teams.

At the same time, Generative AI is increasingly used in software engineering, but BA-specific support is still limited—especially for structured elicitation, ambiguity resolution, and ethical safeguards. This research therefore focuses on **augmenting** (not replacing) BAs in Sri Lankan SMEs through:

- **Adaptive prompt engineering methodologies** that improve requirements elicitation through iterative refinement;
- **Ethical prompting and bias-reduction checks** to improve transparency and trust; and
- **Hybrid AI agents** (human-in-the-loop) that support BA workflows and Agile collaboration.

The work will be designed and evaluated to produce measurable improvements in BA task quality and efficiency.

1.2 Problem Definition

The core problem is the **poor fit** of current AI use for BA processes, which can create inefficiencies, inaccurate artifacts, and ethical risks. Specifically:

1. **Ambiguity in requirements:** Requirements often include vague terms and missing constraints; generic prompting may miss key clarifications, producing incomplete or misleading statements.
2. **Ethical risk and low transparency:** AI outputs can reflect biased phrasing or unspoken assumptions; without explicit checks, trust and adoption remain low.
3. **Alignment and handover friction:** The communication gaps during handover from stakeholder intent to developer-ready artifacts increase rework and delivery delays.

Research Questions:

Main RQ: How can AI-enhanced methodologies and hybrid agents improve BA work in software development while ensuring ethical safeguards and preserving the BA’s human role?

- **SQ1 (Elicitation):** How well do adaptive prompt workflows reduce ambiguity and improve requirement clarity/completeness?
- **SQ2 (Ethics):** Which ethical prompting and bias checks best improve transparency and reduce bias in BA artifacts?
- **SQ3 (Hybrid workflow):** How much does a human-in-the-loop hybrid agent prototype improve BA efficiency and BA–developer alignment in Agile SMEs?

1.3 Objective of the Research

The primary objective is to **develop and evaluate AI-enhanced approaches** that improve BA efficiency, artifact quality, and ethical practices in software projects.

Secondary objectives:

1. **Adaptive prompting:** improve elicitation and refinement using iterative clarification and structured templates.
2. **Ethical prompting:** add lightweight bias/assumption checks and require explicit rationale for AI-generated content.
3. **Hybrid AI agents:** prototype a human-in-the-loop workflow to support BA tasks and BA–developer alignment.
4. **Evidence and guidance:** evaluate using quality rubrics and time/effort measures, and produce adoption guidance for SMEs.

Expected impact: Reduced rework and misalignment, improved BA productivity, and practical guidance for responsible AI adoption in SME software teams.

Chapter 2

Literature Review and Related Work

2.1 Preliminary Literature Review

A preliminary review of recent sources (2023–2026) indicates that generative AI is increasingly discussed in software engineering [8], but **BA-focused coverage remains limited**. Prior work often reports productivity gains (e.g., drafting and summarization) yet provides less guidance for BA-centered elicitation and ambiguity resolution with ethical safeguards.

Studies commonly assess artifacts such as **user stories**, **SRS sections**, and **acceptance criteria** [4, 5]. However, gaps remain in (i) structured clarification workflows, (ii) explicit transparency about assumptions/uncertainty, and (iii) practical bias-aware checks for stakeholder-facing documents [7, 9, 10].

Overall, BA-oriented evaluation in SME settings and sustained **human-in-the-loop refinement** is still underexplored [3, 12]. These gaps motivate this study in Sri Lankan SMEs.

2.2 Related Work

This proposal builds on three directions in software engineering and human–AI interaction:

- **Prompt engineering:** structured prompts and iterative refinement for higher-quality requirements drafts [8, 11].
- **Agentic workflows:** draft–critique–validate decomposition to support BA activities [6, 13].
- **Responsible AI:** bias and hallucination risks motivating transparency and fairness checks [9, 10].

RE (Requirements Engineering) guidance emphasizes clarity, completeness, and testability (rubric-based evaluation), while BABOK aligns with iterative elicitation and validation [3–5].

Novel contribution of this proposal: integrate these strands into a single BA-oriented workflow for SME requirements work by combining an **elicitation loop** (clarification + refinement), a lightweight **ethics/transparency layer**, and a **human-in-the-loop agent** supporting BA–developer alignment [3, 4].

Chapter 3

Methodology

3.1 Materials and Data Sources

This study will use a small, practical mix of primary and secondary data suitable for SME settings:

- **Primary data (Sri Lankan SMEs; target 15–25 BAs):** short surveys and semi-structured interviews on BA workflows, requirements challenges, and AI readiness; optional anonymized artifacts (user stories, SRS fragments, change requests) where permitted.
- **Secondary/public artifacts:** public or simulated requirements cases to ensure evaluation can proceed if organizational access is limited.
- **Data handling:** informed consent, de-identification, secure storage, and exclusion of sensitive/identifying client data.

3.2 Proposed Method

The solution is designed as three integrated layers:

1. **Adaptive prompting workflow:** capture initial stakeholder intent, generate targeted clarification questions, and iteratively refine outputs into user stories, acceptance criteria, and short SRS sections.
2. **Ethical prompting and transparency:** require explicit assumptions/uncertainty notes, traceability to provided inputs, and inclusive-language / stakeholder-coverage checks before finalizing artifacts.
3. **Hybrid (human-in-the-loop) agent:** the agent drafts and critiques, while the BA reviews, edits, and approves any stakeholder-facing deliverables.

3.3 Evaluation

Tasks: transform elicitation notes into user stories and acceptance criteria; resolve ambiguity using targeted clarification questions; update change requests and related artifacts; and produce short, developer-ready handover summaries.

Measures:

- **Requirement quality:** 1–5 rubric for clarity, completeness, and testability [4, 5].
- **Transparency:** assumptions + uncertainty notes + traceability links to source input.
- **Bias/inclusivity:** inclusive-language flags and a stakeholder-coverage checklist [9].
- **Efficiency:** time-on-task and number of rework iterations.

Study design: small A/B-style comparisons where feasible (manual BA workflow vs. AI-augmented workflow) plus qualitative feedback (usefulness, trust, adoption barriers) with thematic analysis.

3.4 Time Plan

The initial plan spans approximately 8 months and may be revised based on access to participants and early findings.

Table 3.1: Tentative time plan

Month	Major Tasks and Subtasks
1	Detailed literature review; finalize research questions; confirm evaluation metrics and instruments (survey/interview guides).
2	Design adaptive prompting methodology; draft ethical framework; architecture design for hybrid agents; prototype wireframes.
3–4	Data collection: recruit 15–25 BAs/SMEs; conduct interviews and surveys; curate/anonymize artifacts; prepare datasets.
5–6	Prototype implementation (web-based toolkit); integrate prompt engine + ethics checks; pilot testing; iterative refinement.
7–8	Full evaluation (quantitative + qualitative); analysis and interpretation; write thesis/proposal updates; prepare paper/report for submission.

Risks and mitigation:

- **Limited data access:** use public/simulated artifacts and keep the prototype evaluation task-based.
- **Low participation/availability:** recruit via professional networks and schedule short, structured sessions.
- **Model limitations or unsafe outputs:** keep a human-in-the-loop review, log assumptions/uncertainty, and avoid sensitive or identifying data.
- **Tool adoption risk:** iterate early with pilot feedback and keep the workflow lightweight for SME teams.

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